Microsoft



SPARROW Mini Assembly & Setup Guide

Solar-Powered Acoustic and Remote Recording Observation Watch

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V1.0

SPARROW Mini Setup Guide

This comprehensive guide is designed to help you seamlessly prepare your SPARROW Mini device for deployment. It's divided into three sections: software setup, hardware setup and sensor setup. By following the step-by-step instructions, you'll ensure a smooth, efficient, and trouble-free setup experience.

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1. Introduction

The biodiversity of our planet is rapidly declining. We have likely reached a tipping point where it is essential to use every tool at our disposal to help preserve what remains[1]. **SPARROW Mini** (Solar-Powered Acoustic and Remote Recording Observation Watch), developed by Microsoft's AI for Good Lab, is a compact, AI-powered edge computing platform designed to operate autonomously in some of the world's most remote environments. Solar-powered and built around an ultra-low-power hardware architecture, **SPARROW Mini** integrates with camera traps, acoustic sensors, and other environmental monitoring devices to collect biodiversity data. Using advanced PyTorch-based wildlife AI models, detections are performed directly on the edge, significantly reducing the amount of data that must be transmitted.

To overcome the communication challenges of remote deployments, **SPARROW Mini** incorporates an integrated **Digi XBee 868 MHz radio (EU) or Digi XBee Pro 900 (US)**, enabling devices to form a self-healing **DigiMesh network**. Individual **SPARROW Mini** units can relay detections, images, sensor measurements, and status information between neighbouring nodes over distances of up to approximately 10 km^2 in open terrain, allowing monitoring networks to be extended well beyond the coverage of cellular or satellite communications. Through multi-hop communication, data from multiple **SPARROW Mini** devices is forwarded across the mesh to a central **SPARROW gateway,** which provides the backhaul connection via 4G cellular or low-Earth orbit satellite communications. This architecture enables large-scale biodiversity monitoring across remote landscapes while maintaining low power consumption, reducing communication costs, and increasing deployment flexibility, allowing researchers to access fresh, actionable insights in real time, no matter where they are.

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Printing Recommendation:
For clarity and accuracy, this guide should be printed in **color**. Many diagrams and component labels **rely on color coding for correct identification**. Printing in black and white may result in unreadable or misleading instructions, increasing the risk of assembly errors.

1.1 Open-Source Statement

The SPARROW Mini device is an open-source, end-user assembled system. This project is provided as-is, without certification or approval for any jurisdiction.

Because components may vary based on user selection, it is the sole responsibility of the end-user to ensure that:

- All hardware components used comply with applicable local regulatory requirements, including but not limited to:
  - Electrical safety standards for wiring, connectors, and power systems.
  - Radio equipment type approval for any wireless modules or transmitters.
  - Environmental and material compliance (e.g., hazardous substances restrictions).

- The completed assembly meets national and regional certification requirements for the location where the SPARROW unit will be installed.

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Commented [BD4R3]: Solved

[1] Hooper, D. U., Adair, E. C., Cardinale, B. J., Byrnes, J. E. K., Hungate, B. A., Matulich, K. L., Gonzalez, A., Duffy, J. E., Gamfeldt, L., & O'Connor, M. I. (2012). A global synthesis reveals biodiversity loss as a major driver of ecosystem change. Natur e, 486(7401), 105–108. https://doi.org/10.1038/nature11118

[2] This value is just an estimation, actual performance varies depending on terrain, antenna configuration, vegetation, interference and local conditions.

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- Any modifications or substitutions made do not violate local laws or safety regulations.

Failure to ensure compliance may result in legal liability, safety hazards, and operational restrictions. Users should consult relevant regulatory authorities and standards before installation. End-users are strongly advised to seek professional review and guidance before installation or operation to confirm compliance with all applicable laws and standards.

Users follow this guide at their own risk. Microsoft provides these instructions for informational purposes only. Certain steps, such as custom wiring, power configuration, or integration of third-party components, may void manufacturer warranties on individual parts. All risks associated with component selection, installation, and usage rest solely with the end-user. Users are strongly advised to review original component warranties and consult qualified professionals before proceeding.

2. Important Safety Information

⚠ CAUTION: Read Assembly & Setup Guide Before Beginning Work

Read this SPARROW Mini Assembly & Setup Guide ("Setup Guide") in its entirety before beginning work. Pay special attention to the safety warnings and instructions in the Setup Guide.

Building and/or repairing any electronic device can present a risk of electric shock, fire, serious personal injury, death, damage to the device or other property, environmental harm, and/or other hazards. Exercise caution when undertaking the activities described in this Setup Guide.

The activities identified in this Setup Guide should only be undertaken by technically inclined individuals with the knowledge, experience, and specialized tools required to construct an electronic device.

⚠ WARNING: Follow all Safety and Health Information

Read and follow all warnings, precautions, handling, and use instructions provided by the manufacturers of hardware components listed in the Hardware Bill of Materials ("BOM"). Failure to follow the manufacturers' instructions provided for set up, use, and care of the hardware components selected for your build can increase the risk of serious personal injury, death, property damage, and/or environmental harm. You can request additional information and support by contacting the hardware component manufacturer.

In addition, please read and follow the additional safety and health information provided in this Setup Guide. Failure to follow the additional information provided can increase the risk of serious personal injury, death, property damage, and/or environmental harm.

⚠ WARNING: Use Personal Protective Equipment

We recommend wearing protective eyewear and gloves when constructing, installing, disassembling, and/or re-assembling a SPARROW Mini unit.

Review and follow the instructions in this Setup Guide when installing, servicing, or inspecting the SPARROW Mini device at heights.

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⚠ WARNING: Battery Safety

Improper use or handling of devices or their batteries may result in battery failure; electric shock; fire; serious personal injury; death; damage to the device, other property, and/or the surrounding environment.

- Before beginning SPARROW Mini unit construction, ensure your workspace is free of flammable debris or materials, has adequate ventilation, and that you have a fire suppressant device (example: fire blanket, container of sand, Class B fire extinguisher) within easy reach or you are within 20 feet of a fireproof enclosure.
- Keep your workspace clear of debris, extra tools, and sharp objects. Avoid using tools that conduct electricity.
- Exercise caution when using sharp tools near the battery to avoid impacting or poking the battery.
- Do not puncture, impact, strike, bend, or crush the battery. If a battery is impacted, dropped, dented, or punctured during installation, DO NOT proceed. Discard the battery pack and use a new one.
- Do not leave loose screws or parts inside the SPARROW Mini unit.
- Do not heat, puncture, mutilate, or dispose of the SPARROW Mini unit batteries installed in your unit in fire. Do not leave or charge the SPARROW Mini unit batteries in direct sunlight or expose its batteries to temperatures outside the recommended operating range identified by the battery manufacturer for an extended period.
- Do not modify or remanufacture the battery installed for your SPARROW Mini unit; attempt to insert foreign objects into the battery, immerse or expose the battery to water or in other liquids or expose the battery to fire, as this might lead to explosion, or other hazards.
- Protect the SPARROW Mini unit and its batteries from excessive rain, snow, or other types of moisture.
- Ensure you handle, recycle, and/or dispose of used or damaged batteries in accordance with local laws and regulations.

⚠ Electrical and Wiring Safety

Always use the proper wire gauge when constructing, installing, or servicing the SPARROW Mini device. Failure to use the proper wire gauge may result in overheating and fire hazards due to undersized wiring, electrical shock or short circuits from improper connection, device malfunction or permanent damage; follow applicable safety, code and compliance standards.

- Follow the wire gauge specifications provided by the component part manufacturer.
- Do not substitute smaller or unapproved wire sizes or types.
- Ensure all connections are secure and insulated to prevent accidental contact.
- Avoid working on electrical installations during adverse weather conditions such as high winds, thunderstorms, rain, ice, or snow.
- We recommend that all electrical wiring-related activities only be conducted by a trained and licensed professional using approved materials.

To help protect your SPARROW Mini unit from environmental hazards and wildlife interference, it is strongly recommended to install any exposed wiring inside approved conduits to prevent physical damage from rodents, birds, or other animals; moisture intrusion that can lead to

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corrosion or short circuits; or UV degradation of cables in outdoor environments all of which can expose powered wires and pose an electric shock and fire hazard though short circuits.

Always select conduits rated for outdoor use and compliance with local electrical codes. Secure the conduit properly to avoid strain on connections and maintain system integrity.

⚠ WARNING: Working at Heights

Do NOT work at heights without proper safety measures and use of personal protective equipment when installing, servicing, or inspecting the SPARROW Mini device at height. Failure to comply may result in falls leading to severe injury or death, damage to equipment, surrounding property, and/or the environment, and violation of workplace safety regulations.

- Always use approved fall protection equipment (harnesses, anchors, guardrails).
- Ensure ladders, scaffolds, and platforms are stable and secure before use.
- Do not work alone—maintain communication with a safety partner.
- Avoid working at heights during adverse weather conditions such as high winds, thunderstorms, rain, ice, or snow.
- Follow all local occupational health and safety standards.

⚠ WARNING: Proper Installation

To avoid hazards related to improper device installation, installation must be performed by individuals who have read this Setup Guide and fully understand the installation requirements and environment prior to beginning work.

Consult a professional installer if you do not have the necessary equipment or expertise, or if you are uncertain your proposed mounting environment and mounting solutions will properly support your SPARROW Mini device and not create a hazard, damage, or harm to the installation area or structure.

⚠ Safe Ground Anchoring

For ground mounted units, ensure the SPARROW Mini device is securely anchored to stable ground before installation or operation. Improper anchoring may result in device tipping or collapse, causing severe injury, death, or other damage. Insufficient anchoring can lead to system misalignment, reducing performance and reliability and create hazards during high winds, seismic activity, or accidental impact.

- Use approved anchoring hardware and methods as specified by local requirements.
- Verify soil stability and load-bearing capacity before installation.
- Do not install on loose, uneven, eroding, or flood-prone surfaces without adhering to regulations and requirements for this area.
- Inspect anchors regularly for signs of corrosion, loosening, or structural compromise.
- Follow all local building codes and safety regulations for ground-mounted systems.

Failure to properly anchor the unit can lead to catastrophic failure, personal injury or death, and legal non-compliance.

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⚠ WARNING: Avoid Flammable or Explosive Environments

Do not build, install, or operate the SPARROW Mini unit near flammable materials or within a flammable or explosive environment. Keep all flammable liquids, gases, and solids away from the installation site. Do not install the SPARROW Mini unit in areas with combustible dust, vapors, or fumes. Failure to comply may result in fire or explosion hazards due to the ignition of combustible substances, causing severe injury or death to users, bystanders, and wildlife, property damage, and/or permanent damage to the SPARROW Mini unit and the surrounding environment.

⚠ WARNING: Unseen Hazards

Mounting locations can contain unseen hazards or obstacles. Cutting or drilling into an unseen hazard may cause serious personal injury, death, or other harm.

Assess your mounting environment and always make sure there are no unseen hazards associated with the mounting surface prior to drilling and/or cutting. It is the installer's responsibility to locate unseen hazards prior to and to avoid any identified hazards during installation.

⚠ WARNING: Ergonomic Lifting

To reduce the risk of lifting related injuries, death, or damage to your SPARROW Mini unit or the environment, use good ergonomic lifting practices during installation, maintenance, and use, including but not limited to:

- Plan ahead. Make sure the lifting team agrees on the plan.
- Determine if you can lift the unit. Is it too heavy or too awkward?
- Decide if you need lifting aid.
- Check your environment for obstructions and slippery surfaces.
- Lift with your legs, not your back.
- Bend at your knees, keeping the back straight.
- Keep the unit close to your body.
- Center your body over the unit. Keep the feet about shoulder width apart.
- Lift straight up smoothly.
- Keep your torso straight; do not twist while lifting or after the load is lifted.
- Set the unit slowly and smoothly into place a straight back; do not let go until the unit is set in place.

⚠ WARNING: Solar Panel Glass

Do not install or use the SPARROW Mini unit if any solar panels are cracked, broken, or otherwise damaged. To reduce the risk of personal injury, avoid touching a solar panel that is broken, chipped, or cracked and arrange to have it replaced. Using damaged solar panels may result in electrical hazards, including shock or short circuit, fire risk due to exposed or damaged internal components; and reduced system performance or failure.

⚠ WARNING: Do not Modify or Alter the Solar Panel or Battery Terminals

To reduce the risk of fire, electrical shock, causing severe injury or death to users, bystanders, wildlife, the environment, and/or the SPARROW Mini unit, do not modify, tamper with, or attempt

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to alter the DC terminals of the solar panel array or the terminals of the battery bank used to construct the SPARROW Mini unit.

3. Prerequisites

Before you begin, ensure you have **all the necessary hardware listed in the Bill of Materials (BOM) and software components ready**.

How to Use This Guide

Before starting any assembly, **read each section completely**, including all steps and notes—before starting the build. This ensures you understand the sequence, required tools, and safety precautions. Skipping ahead or acting without reviewing the entire section may lead to errors, damage, or safety hazards.

Assembly Time Estimate
Please note that actual build time may vary based on experience. As a guideline, **allow 20 Minutes**.

Power Requirements
Ensure your build space has access to at least **two electrical outlets**. This is necessary to power both the PC (used for flashing and configuration) and the Pi Zero 2 W in parallel during setup. Additional outlets may be required for charging tools or other peripherals.

3.1 Software Requirements:

- **Host PC for running Raspberry Pi Manager**
- **Internet Connection** (for downloading necessary files)
- **SPARROW Mini Setup Script** (this script can be found in the SPARROW Mini GitHub Repo: https://github.com/microsoft/sparrow-client)

3.2 Hardware Requirements:

- **Raspberry Pi Zero 2 W**
- **SD card**
- **Power Supply: Compatible with Raspberry Pi Zero W2**
- **Host PC with internet access**
- **SD Card reader**
- **Ethernet connectivity**

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4. HARDWARE SETUP

4.1 Introduction

The SPARROW Mini device can be divided into 3 main component systems:

- **COMPUTE**: Includes the solar controller, edge device and associated devices and the XBee radio.
- **POWER**: Solar panel.
- **SENSOR**: Includes Peripheral Sensors (Cameras, Microphones, External Sensors, etc.) that connect to the SPARROW Mini unit.

The following steps will guide you through the assembly process of a standard SPARROW Mini using solar energy. The guide will cover the recommended components and describe how the components interact with each other. Following this guide may assist users in assembling and configuring a SPARROW Mini unit; however, successful assembly, operation, performance, safety, and regulatory compliance depend on a variety of factors, including component selection, assembly quality, installation environment, and maintenance.

4.2 Component Inventory

The following components highlight each of the required parts to build a full SPARROW unit. This guide provides a brief explanation and description of the function of each component and how it works in the overall system.

You can download the complete SPARROW Mini Bill of Materials from:
aka.ms/sparrowminibom

Voltaic 20w Solar Panel and Battery:



The 20 Watt 6 Volt solar panel is lightweight, waterproof and easily mountable for long term outdoor applications. SPARROW Mini is powered by a single Voltaic battery which contains a built-in solar controller. The V75 USB Battery Pack features 20,100mAh and is designed to power smartphones, tablets, cameras as well as IoT devices and microcontrollers.

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Edge Computing Device:



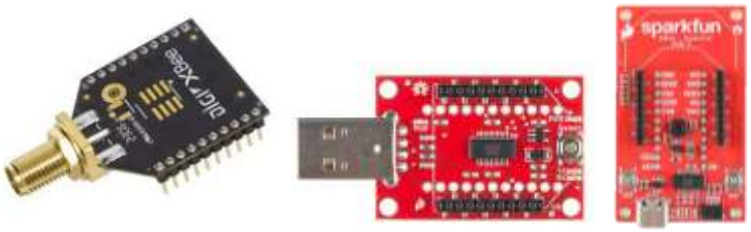
The SPARROW Mini uses a Raspberry Pi Zero 2 W. This component runs all the required software services including the AI models. All the sensors directly interface with the Raspberry Pi to provide real-time environmental readings.

Voltaic Solar Panel Mount (Optional):



Securely mounts the solar panel and SPARROW Mini. Can be used with a strap or can be physically screwed into a fixed anchor.

XBee RF with UART to USB adaptor:



XBee 898/900 is a compact XBee RF module add-on board for long range low-power wireless communication applications. The module can run either a proprietary DigiMesh or point-to-multipoint networking protocol with reliable long range performance and, data redundancy. The **UART to USB adaptor** can be either USB A or USB C, any of these will work.

915MHz LoRa XBee Antenna:

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High-gain 915 MHz XBee/LoRa antenna designed to extend wireless communication range and improve signal reliability in outdoor deployments.

USB Ethernet Adapter:



The USB ethrnect addapter is used to connect the USB MOSFET and the XBee UART USB adapter.

USB Mosfet Power Switcher (Optional):



This module enables power to a USB device to be turned on/off via a USB power switch (MOSFET). Power goes IN via the micro-USB side and OUT via the USB-A side.

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Electrical Junction Box (Optional but Recommended):



This electrical junction box is IP65 rated, or NEMA 4(x) equivalent rating, making it suitable for outdoor applications and able to withstand rain, dust, and other outdoor elements.

Dupont Cables (Optional):



Dupont cables are used to control the USB Mosfet Power Switcher via the Pi's GIPO port.

SDWire3 (optional):



The SDWire3 board switches an SD/microSD card between a sensor and the SPARROW Mini, allowing the Raspberry Pi to access and download data stored on the sensor's card.

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Micro USB to USB A cable:



Micro USB cable used for connecting external devices to the SPARROW Mini.

USB C to USB A cable:

4.3 SPARROW Hardware Assembly Guide

4.3.1 **COMPUTE**: Assembling the Raspberry Zero W

1. Gather Required Tools and Components

- Raspberry Pi Zero 2W
- USB Ethernet Adapter
- Raspberry Pi Zero 2W compatible power bank
- XBee RF board
- UART USB adapter
- XBee compatible Antenna
- Micro USB to USB A Cable

2. Power Down and Prepare the Workspace

- Shut down the Raspberry Pi Zero 2W completely.
- Disconnect the power supply and all peripherals (USB, HDMI, etc.).
- Work on an anti-static mat or non-conductive surface to avoid electrostatic discharge.

Mounting the Pi Zero 2W with Velcro or Double Sided tape:

Tools & Materials you will need for this section:

- Raspberry Pi Zero 2W
- Weatherproof Outdoor Extension Cord Cover Box or Electrical junction box with perforated backplate
- Heavy-duty zip ties (Optional)
- Drill (if holes need enlarging) (Optional)
- Marker or pencil
- Double Sided Tape
- Ready to Stick Velcro (Optional)

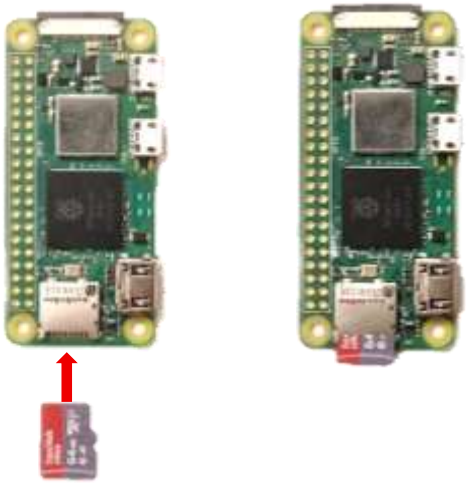
Step by Step Guide:

1. Prepare the workspace.

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2. If you already have an imaged Micro SD Card, you can insert it now into the Micro SD card slot of the Raspberry Pi Zero 2W Micro SD slot as shown below. If not, you can carry on and complete the build and image the SD card after the complete system assembly is completed.

a) b)



3. Connect the USB Ethernet Adapter into the Micro USB port of the Raspberry Pi Zero 2W labeled “USB” as shown below:



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4. Using a small square of double-sided tape, stick and secure the Raspberry Pi Zero 2W board to the body of the USB Ethernet Adapter.
   IMPORTANT: make sure that the USB ports are facing the correct direction so that when connecting the UART to USB board the pins for the XBee RF board face upward.



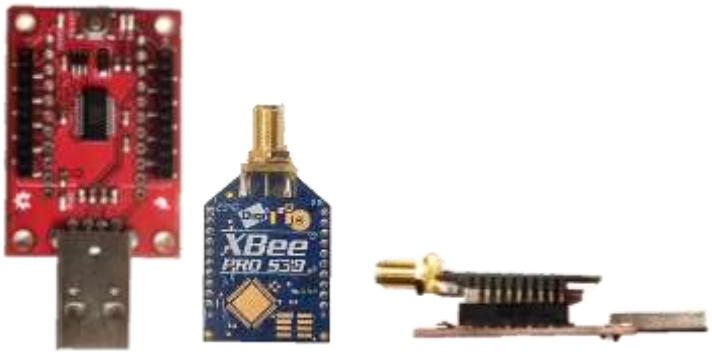
   PRO TIP: You can tell which side of a USB-A female port is the correct one by looking at the solid plastic block inside.
   - Inside the Port (The Plastic Block)
   - Look inside the metal opening with a flashlight.
   - You will see a small white plastic tongue or block inside.
   - The side that has the open space **below** the plastic block is the top and should be facing up for our use case.

5. At this point the assembly should look as follows:



6. Connect the XBee RF board onto the USB-UART adapter board by aligning and pressing in the pins into the socket as shown below, in this case we will use the USB-A version of the UART adapter to save space.

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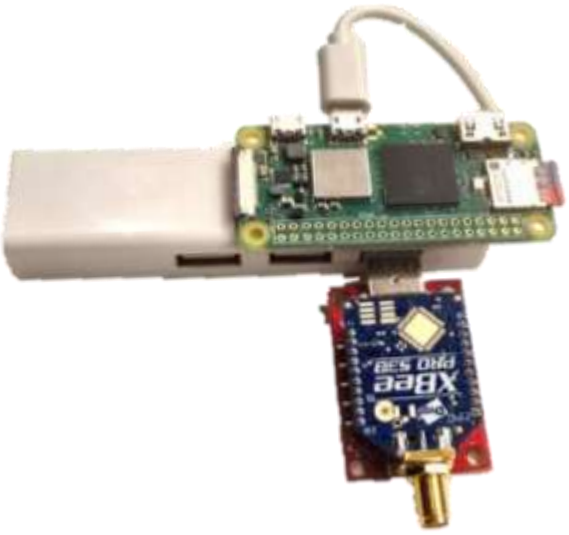


Important Note: The USB-C version works just the same as the USB-A but will require an extra USB-C to USB-A cable. The USB-C version allows you to freely position the board inside the enclosure but does take up more space since it requires the extra USB-C Cable.

7. This is how your finished XBee assembly should look like:



8. Insert the XBee assembly into one of the USB-A ports of the USB Ethernet Adapter as shown below:



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9. Connect the USB-A to Micro USB cable to one of the output ports of your battery bank as shown below:



10. Connect the Micro USB cable coming from the battery bank to the Micro USB port in the Raspberry Pi Zero 2W labelled “PWR IN”. At this point you should see the Raspberry Pi Zero 2W and the XBee board assembly power up indicated by the lights powering up.
11. Secure the USB Ethernet – Raspberry Pi Zero 2W assembly onto the battery by using a small piece of double-sided tape.
    The resulting assembly at this point should look like this:



12. At this point connect the XBee antenna using the RP-SMA antenna connector on the XBee chip:

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PRO TIP: For maximum extended range, you can swap this antenna for a high gain external antenna like the one shown below:



13. Finally continue by connecting the Solar panel 5V terminal into the battery bank as shown below:

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14. Congratulations! the main Sparrow Mini assembly is now complete!

15. In order to protect the components from the environment, a weatherproof enclosure is recommended. Utilizing IP65, or Nema 4X equivalently rated weatherproof enclosures like the one shown below is highly recommended.



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16. To install the system into the enclosure simply route the solar panel cable through the rubber seals at the bottom of the enclosure and procede to secure the complete Battery+Raspberry Pi Zero 2W+XBee assembly to the back plate, by using ready-to-stick Velcro tape or VHB (Very High Bond) double-sided tape, or by simply using zip-ties. The finished system should look like this:



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17. If you are using the recommended Voltaic solar panels and battery packs, and have purchased the recommended solar panel mounting plate, then procede to secure the solar panel onto the mounting plate, you can use small zip ties or their screw mouting kit through the commonly aligned holes. The weatherproof enclosure can also be zip-tied onto the mouting plate under the solar panel, or affixed below the solar panel when deployed.



18. Picture of a field deployed unit using one of the extended range antennas:



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IMPORTANT: You do not need to understand the individual software components. The installer configures and starts them automatically.

5. Software Installation and configuration

There are only two transport modes to understand:

Mode	When to use it	What the SPARROW Mini does
RELAY / DigiMesh	Remote/off-grid site with no internet at the SPARROW Mini.	Sends detections over XBee. Other SPARROW Mini nodes can relay the data until it reaches the SPARROW Gateway, which has cellular or Low Earth Orbit satellite backhaul.
DIRECT	The SPARROW Mini has Wi-Fi or 4G internet.	Sends detections directly to the SPARROW server. XBee is not required for forwarding.

RECOMMENDED: For remote XBee 868 DigiMesh deployments, select RELAY mode.

5.1 Prepare the Raspberry Pi

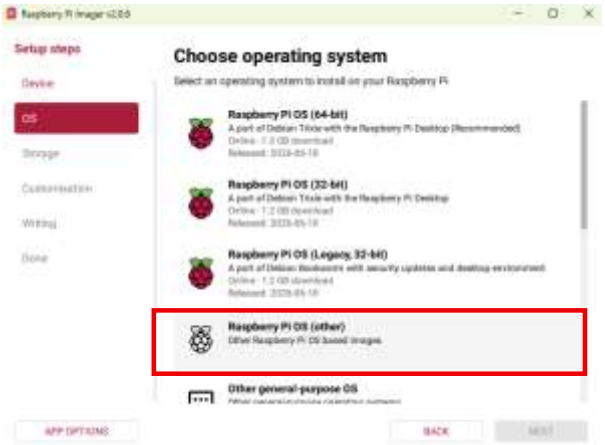
It is recommended you complete the hardware build first. Then prepare the microSD card:

1. Download and Install Raspberry Pi Imager on a PC and insert the microSD card (https://www.raspberrypi.com/software/).
2. Select Raspberry Pi Zero 2 W.

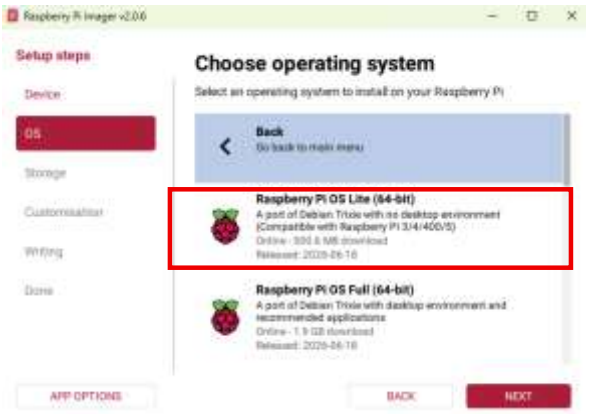


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3. 3. Select Raspberry Pi OS (Other) and click next:



4. 4. Select Raspberry Pi OS Lite (64-bit).



5. Set the primary username to sparrow.
6. Configure temporary Wi-Fi and a secure password for the initial setup.
7. Write the card, safely eject it, and install it in the SPARROW Mini.

IMPORTANT: The Raspberry Pi username must be sparrow.

5.2 Run the SPARROW Mini Configuration Script

Power the SPARROW Mini and log in to the Raspberry Pi. You only need the setup script on the Pi. When it runs, it downloads the SPARROW Mini software repository automatically, installs the required software, configures the device, and starts the service.

Run the SPARROW Mini setup script:

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sudo ./install.sh --pass <SPARROW-MINI-WIFI-PASSWORD> --pat <GITHUB-TOKEN>

The installer will guide you through the setup. When asked which XBee family is fitted, select 868 for the XBee SX 868.

Replace <SPARROW-MINI-WIFI-PASSWORD> with the password you want to use for the SPARROW Mini local Wi-Fi hotspot.

The installer automatically:

- Installs the required software.
- Creates the local SPARROW Mini Wi-Fi hotspot.
- Configures the XBee 868/900 radio based on selection.
- Installs and starts the SPARROW Mini service.
- Starts the node in RELAY / DigiMesh mode by default.

IMPORTANT: If the repository is still private, GitHub access may be required during installation. This is only an installation requirement.

5.3 Open the SPARROW Mini Status Page

After setup, field checks can be done from a phone, tablet, or laptop. SSH is not normally required.

1. Connect your phone or laptop to the SPARROW Mini Wi-Fi hotspot.
2. Open a web browser.
3. Go to http://10.42.0.1:8080.
4. Confirm that the SPARROW Mini status page loads.

The page shows the information a field technician normally needs:

- XBee signal/link status in RELAY mode.
- Cloud connection status in DIRECT mode.
- Image and audio detector status.
- Files waiting to be sent.
- Basic device configuration and system status.

6. XBee DigiMesh Setup

The XBee is normally configured during installation. To configure it again:

1. Connect to the SPARROW Mini Wi-Fi hotspot.
2. Open http://10.42.0.1:8080.
3. Open the menu.
4. Select Configure XBee.
5. 5. Wait for configuration to finish and check the XBee link status.

IMPORTANT: All nodes in the same mesh must use compatible radios in the same frequency family.

7. RELAY / DigiMesh Mode

Use RELAY mode when the SPARROW Mini is deployed away from cellular or satellite connectivity.

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1. The camera or recorder creates a capture.
2. The SPARROW Mini runs AI locally.
3. Empty captures can be discarded to save power, radio airtime, and storage.
4. Useful detections are sent over the XBee radio.
5. If the Gateway is not directly reachable, another SPARROW Mini can relay the data.
6. The data continues through the mesh until it reaches the SPARROW Gateway.
7. The SPARROW Gateway forwards the data using cellular or Low Earth Orbit satellite backhaul.

KEY POINT: A SPARROW Mini can monitor its own camera while also acting as a relay for neighbouring SPARROW Mini devices.

8. DIRECT Mode

Use DIRECT mode when the SPARROW Mini itself has Wi-Fi or 4G internet.

1. Open the SPARROW Mini status page.
2. Open the menu and change Transport mode to DIRECT.
3. Open Server & Onboarding.
4. Enter the SPARROW server address, API key, and onboarding email.
5. Select Onboard device.
6. Confirm that Cloud Upload shows a successful connection.

IMPORTANT: The capture and AI process does not change between RELAY and DIRECT modes. Only the way the detections are sent changes.

9. Camera and Audio Modes

The same SPARROW Mini can be used for camera monitoring, audio monitoring, or both.

Monitoring	Typical source	What happens
Camera	USB / SDWire / Wi-Fi-IP camera	Images are collected, checked by AI, and useful detections are forwarded.
Audio	USB recorder / USB microphone	Audio is collected, checked by AI, compressed, and useful detections are forwarded.
Camera + Audio	Both	The SPARROW Mini handles both data types and sends them using the selected transport mode.

10. Blank Capture Setting

Blank filtering controls whether captures with no AI detection are retained.

- Blank filtering enabled / keep blanks off: empty captures are discarded.

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- Keep blanks on: empty captures are retained and can be forwarded.

This can be changed from the SPARROW Mini status page for images and audio.

11. Recommended DigiMesh Deployment Spacing

Use the following as simple field-planning guidance. Actual performance depends on antenna height and gain, vegetation, terrain, obstructions, and local radio conditions.

Environment	Recommended node spacing	Guidance
Open terrain	1,500–2,000 m	Keep antennas high, vertical, and as clear of obstructions whenever possible.
Woodland / dense vegetation / terrain variation	1,000–1,200 m	Use shorter hops and position nodes to maintain a strong, reliable link.

Each SPARROW Mini added to the network can extend the mesh and can also operate its own wildlife camera or sensor.

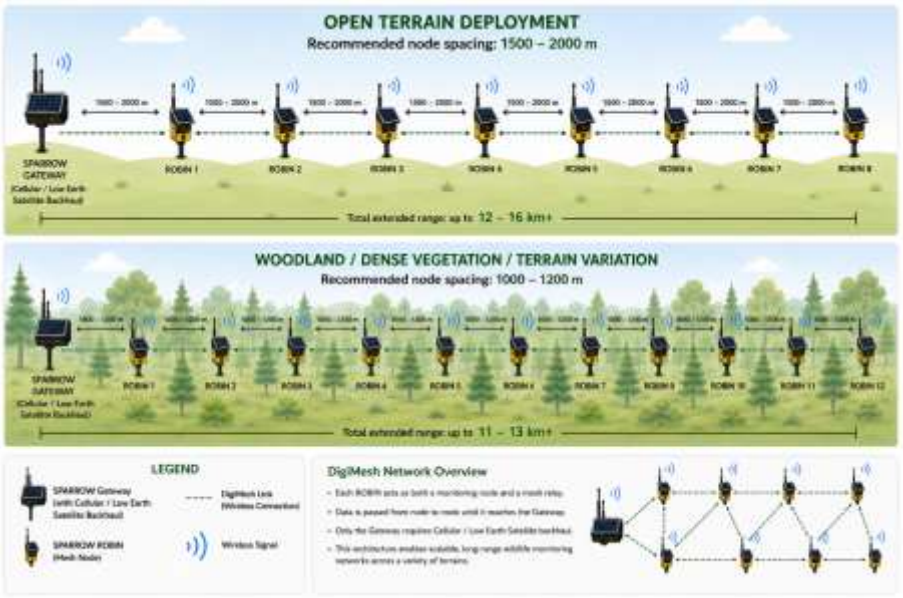


Figure — Example SPARROW Gateway and SPARROW Mini DigiMesh deployment layouts.

12. Final Field Check

- SPARROW Mini powers on correctly.
- Phone/laptop can connect to the SPARROW Mini Wi-Fi hotspot.
- Status page opens at http://10.42.0.1:8080.

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- Correct transport mode is selected: RELAY or DIRECT.
- XBee link is healthy in RELAY mode.
- Cloud connection is healthy in DIRECT mode.
- Camera and/or audio source is operating.
- AI detector shows successful processing.
- No unexpected transmission backlog is building.
- Antenna is secure, vertical, and mounted as high and clear as practical.
- Solar panel, battery, enclosure, cabling, and mounting are secure.

13. SENSOR SETUP

13.1 Configuring ReoLink Wi-Fi Cameras

Once sparrow mini is running you can connect and configure the ReoLink cameras to transmit images to the SPARROW unit.

1. **Download the ReoLink app:**
   - download the ReoLink app on your mobile device from the Google Play Store or Apple App Store (no registration is required to use the ReoLink app):



2. ⚠️ **IMPORTANT: Do NOT log in to any Reolink** account in the app during setup.
   If you are already logged in, log out before continuing.
   This step is critical—if you remain logged in, the cameras will not send alerts when there is no active internet connection.
3. **Connect your phone to the SPARROW Wi-Fi network:**
   - Open your Wi-Fi settings on your phone and search for the SPARROW SSID. Connect to the SSID and entre your password.
4. **Open the ReoLink app and click on the icon shown below to add a new camera:**



5. **Each camera comes with its own distinct QR code which can be found on the outside of the camera. Using the ReoLink app scan the QR code as shown below:**

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6. **Follow the on-screen instructions to join the ReoLink camera to the SPARROW SSID.**

7. **Enter a local password:**



8. **Name the camera and click the button Next to enter the next page:**

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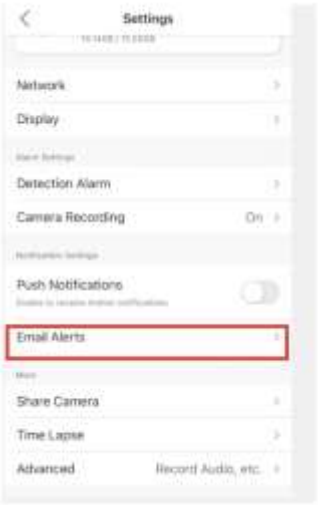


9. **Read the tips and click Use now to finish the Initialisation:**



10. **On the camera settings page click on Email Alerts:**

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11. **Enable Email Alerts then click Set up Now:**



12. **Fill in the required information and click save. You must use the following settings:**
    - Sender address: camera@localnetwork.com
    - Password: (Check the sparrow setup script)
    - SMTP Server 10.42.0.1
    - Port: 25
    - Turn off SSL/TLS
    - Recipient Adress 1: alerts@localnetwork.com (when using Android this setting is available after first setting it up with the Sender Address)
    - Click Save
    - Click Test and make sure you get the test success confirmation message

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13.2 Peripheral Sensor Assembly (OPTIONAL)

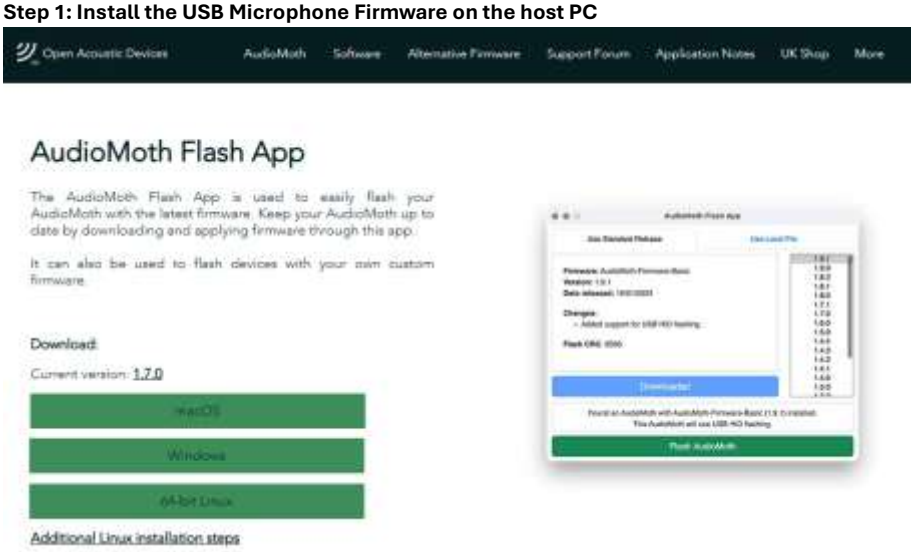
AudioMoth DEV Microphone Setup

What You'll Need:

- AudioMoth DEV Board
- AudioMoth DEV Case
- Micro USB cable
- Cable gland (for the case)
- Internet access for firmware download
- Precision knife

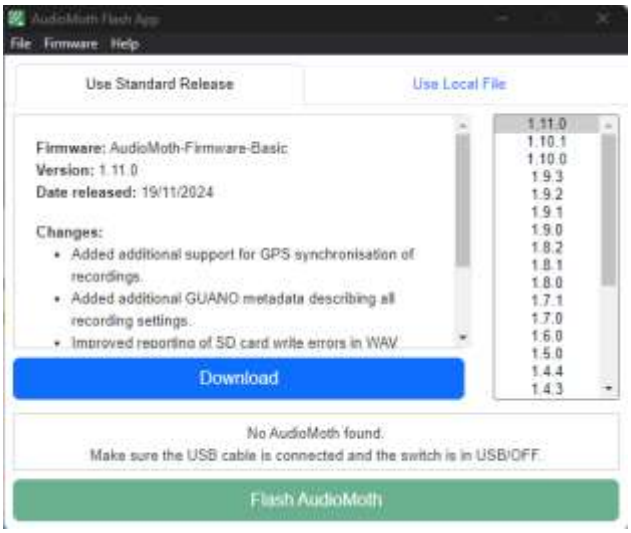
Flashing the AudioMoth (OPTIONAL)

Step 1: Install the USB Microphone Firmware on the host PC

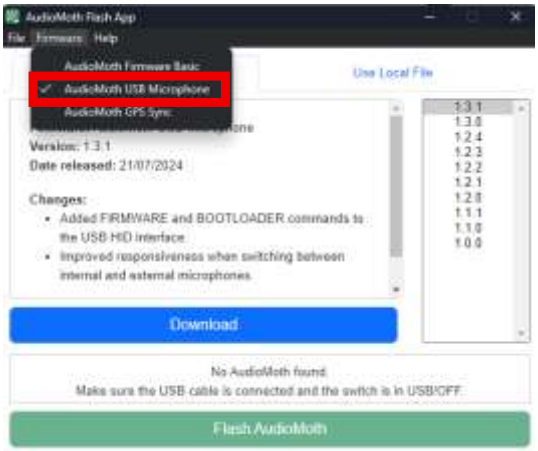


1. **Go to the** [AudioMoth website](#) and download the **AudioMoth Flash App**. The AudioMoth Flash App is used to flash your AudioMoth with the latest firmware.
    a. Download, Install the AudioMoth Flash App on your computer (you can use Windows, macOS or 64-bit Linux systems).
2. **Open the AudioMoth Flash App** on your computer.

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Select the USB Microphone firmware from the Firmware menu.



3. Select the **1.3.1 Firmware** version from the list at the right and click the **BLUE** “Download” Button.

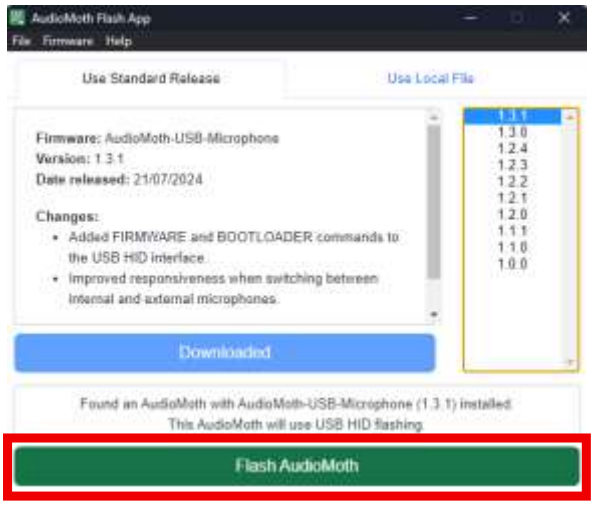
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4. **Connect the AudioMoth** board to the computer using the **Micro USB port**.
5. Make sure the micro-USB cable is connected, and white micro-switch in the AudioMoth is set to the **USB/OFF position**.



6. **Flash the firmware** by clicking the **GREEN** "Flash AudioMoth " button.



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7. After the flashing process is completed the progress bar will be full and a "*Firmware has been successfully updated.*" message is displayed, at this point you can disconnect the micro-USB cable from the AudioMoth board.
8. Flip the white micro-switch on the AudioMoth from the **USB/OFF position** to the **CUSTOM position**.
9. **Your AudioMoth is now ready for deployment.**

Installing the AudioMoth

Step 1: Prepare the AudioMoth DEV Case.



1. **Open the case** and locate the battery cradle inside.
2. **Remove the battery cradle:**
    a. Undo the straps holding it in place.
    b. Lift the entire cradle out of the green box.
    c. You should now have an empty case.

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Step 2: Install the Cable Gland.



1. **Find the cable passthrough placeholder** M16 cap on the case.
2. **Remove the placeholder M16 cap.**
3. **Install a PG9 cable gland** in its place.
    a. Make sure it is tightened securely to prevent movement.

Step 3: Prepare the Micro USB Cable.

1. **Check if the Micro USB connector fits through the cable gland.**
    a. If it does not fit, proceed to the next step.
2. **Trim the Micro USB connector carefully:**
    a. Use a precision knife to shave off some of the outer rubber casing.
    b. Important: Do NOT cut into the internal data wires.
    c. Take your time—small cuts are safer than deep cuts.

Step 4: Route the Cable.

1. **Insert the Micro USB cable through the cable gland** from the outside of the case.
2. **Pull enough cable inside** so it can comfortably reach the AudioMoth board.

Step 5: Connect and Position the AudioMoth.

1. **Plug the Micro USB connector into the AudioMoth.**
2. **Place the AudioMoth board inside the case** in its designated position.
3. **Ensure the cable is neatly routed** and not pinched by the case edges.

Step 6: Final Check.

1. **Confirm the cable gland is tight.**
2. Make sure the AudioMoth is secure in its locking tabs, and the cable is connected properly.
3. Close the case.
4. AudioMoth is now ready for field deployment, use the provided Velcro straps or zip-ties to secure it in its final location.

Commented [LP5]: @Bruno Demuro Segundo @Metaverse LLC - ready for deployment or testing and installation?

Commented [BD6R5]: Field deployment.

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5. Connect the USB-A side of the cable to one of the free USB-A ports in the USB Ethernet Adapter of the Sparrow Mini and route the cable through the pre-installed cable seals.

13.3 LEGACY Sensor Retrofit (OPTIONAL)

13.3.1 USB Storage Interfacing: Access sensor storage via USB

Use this optional assembly when you are planning on using a USB-connected sensor that enters storage mode after its USB port is connected (eg: Camera Trap, Acoustic Recorder, etc). The 8086.net USB Power Switch lets the SPARROW Mini control that connection and pass the sensor data through to the SPARROW Mini or SPARROW unit.

ASSEMBLY OUTCOME The switch controls when the sensor enters USB storage mode while allowing stored files to pass through the USB connection to the SPARROW system.

Case Modification and Weatherproofing

WARNING: The USB cable must pass from the sensor usb port to the 8086.net USB Power Switch outside the sensor housing. Some sensors do not provide enough clearance for the USB cable to pass through. Drilling, cutting, filing, enlarging an opening, modifying housing, or replacing part of the housing may be required. These changes may void the manufacturer's warranty and may reduce the original weather or ingress-protection rating. After routing the USB Cable, install suitable strain relief and reseal the opening using an appropriately rated cable gland, gasket, sealant, or equivalent method. Inspect and test the modified enclosure for water and dust resistance before field deployment. Do not deploy a modified sensor until the housing has been properly resealed and weatherproofed.

For long-term deployments, use a solar-powered camera trap or sensor whenever practical. Confirm that the sensor power system can support the planned capture schedule and that all external power and data entries remain sealed for the deployment environment.

Required Parts and Tools

- Raspberry Pi Zero 2 W or Raspberry Pi Zero 2 WH
- 8086.net USB Power Switch board
- 2 x 20 male GPIO header, required only for a Raspberry Pi Zero 2 W
- Soldering iron, electronics-grade solder, and a suitable holder or jig required only for a Raspberry Pi Zero 2 W or a GPIO pin header press-on kit.
- Three female-to-female DuPont cables: **Black**, **Green**, and **Red** (we will use these colors for the build)
- Micro USB to USB-A data cable, shown in the photos as the **YELLOW** USB cable (the YELLOW cable in this guide was used for representation purposes only, any Micro USB to USB-A cable will do)
- USB cable for the selected sensor, shown as the large **BLACK** USB cable
- White USB hub used in the SPARROW Mini assembly

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GPIO Wiring Reference

Cable Color	USB Switch terminal	Raspberry Pi GPIO Connection	Physical pin
Black	GND	Ground	Pin 6
Green	EN	GPIO17, BCM 17	Pin 11
Red	5V	5V power rail	Pin 2

Safety and Preparation

WARNING: Disconnect all power before soldering or changing GPIO wiring. Confirm every connection before applying power. A misplaced 5V connection can damage the Raspberry Pi, USB switch, hub, or sensor.

Step 1: Power Down the Assembly

1. Shut down the Raspberry Pi Zero 2 W completely.
2. Disconnect the battery or external power supply.
3. Disconnect all USB peripherals.
4. Place the components on a clean, non-conductive and static-safe work surface.

Step 2: Identify the Raspberry Pi Version

Raspberry Pi Zero 2 WH: The 40-pin GPIO header is already installed. Continue to Step 4.

Raspberry Pi Zero 2 W: The GPIO header must be installed before the DuPont cables can be connected. Complete Step 3.

Step 3: Solder the GPIO Header, Pi Zero 2 W Only

5. Insert the 2 x 20 male GPIO header into the 40 GPIO holes. Keep the longer pins above the component side of the Raspberry Pi.
6. Hold the header fully seated, straight, and perpendicular to the board.
7. Solder one pin at each end of the header. Check alignment before continuing.
8. Solder the remaining pins individually, using only enough solder to form a clean joint.
9. Inspect the header for solder bridges, incomplete joints, loose pins, or solder debris.
10. Allow the Raspberry Pi to cool completely before connecting any cables.

Do not solder a Raspberry Pi Zero 2 WH. The header is supplied pre-installed on that model.

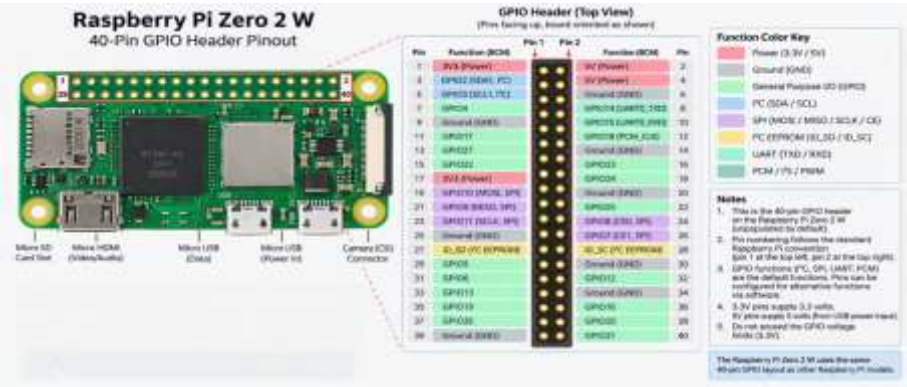
Step 4: Locate the Required GPIO Pins

Find the physical pin numbers on the 40-pin header before attaching the wires:

- Physical Pin 2: 5V
- Physical Pin 6: Ground
- Physical Pin 11: GPIO17

IMPORTANT: Physical pin numbers alternate between the two rows. Verify the pin layout rather than counting along only one row.

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Connect the Control Wiring

Step 5: Connect the DuPont Cables to the Switch

11. Connect the **BLACK** cable to the terminal labeled GND.
12. Connect the **GREEN** cable to the terminal labeled EN.
13. Connect the **RED** cable to the terminal labeled 5V.
14. Check the printed labels on the board. Do not rely on cable color alone.



Figure 1. Switch control terminals and cable colors: black GND, green EN, and red 5V.

Step 6: Connect the DuPont Cables to the Raspberry Pi

15. Connect **BLACK** GND to Raspberry Pi physical Pin 6.
16. Connect **GREEN** EN to Raspberry Pi physical Pin 11, GPIO17.

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17. Connect **RED** 5V to Raspberry Pi physical Pin 2.

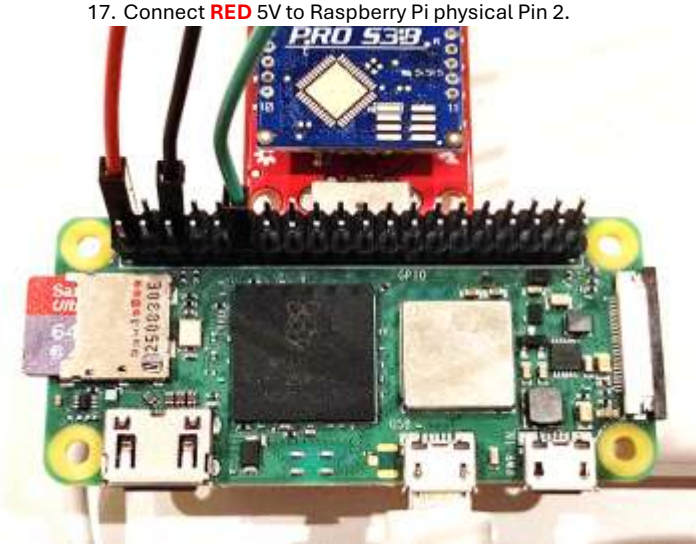


Figure 2. Switch control terminals and cable colors on Pi Zero's GPIO: black GND, green EN, and red 5V.

18. Verify all three connections against the wiring table before restoring power.

CHECKPOINT: The green EN lead is the control signal. The black lead provides the common ground, and the red lead connects the 5V line shown in this SPARROW Mini assembly.

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Connecting the USB Data Path

Step 7: Connect the Switch to the White USB Hub

19. Connect the Micro USB end of the **YELLOW** cable to the Micro USB port on the 8086.net board.
20. Connect the USB-A end of the **YELLOW** cable to an available port on the white USB hub.
21. Confirm that the connectors are fully seated and that the cable does not place strain on the switch board.

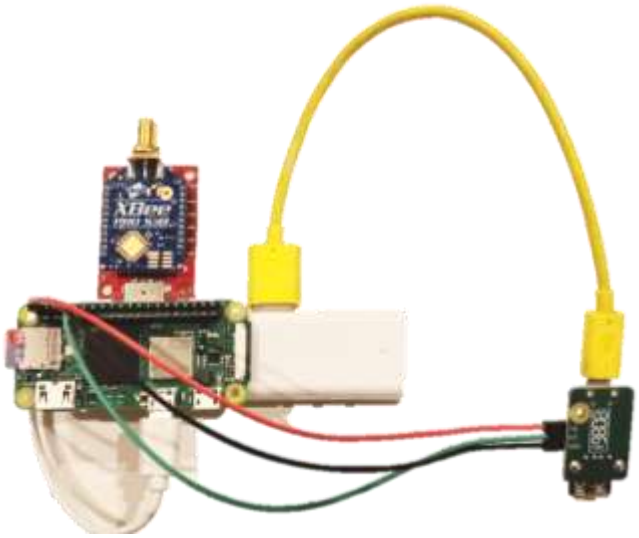


Figure 3. The yellow Micro USB cable connects the switch input to the white SPARROW Mini USB hub.

Step 8: Connect the Sensor

22. Connect the sensor cable to the USB-A female port on the 8086.net USB Power Switch.
23. In the reference photographs, the sensor connection is the large BLACK USB cable leaving the frame.
24. Route and secure the sensor cable so it cannot pull on the switch board, GPIO header, or DuPont leads.

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25. Use a sensor (Camera Trap, Acoustic Recorder, etc) that enters USB storage mode when connected to a USB host.



Figure 4. Completed USB path with the yellow hub cable and black sensor cable connected.

Final Inspection and Functional Test

Step 9: Inspect the Assembly Before Power-Up

- **Black** wire: switch GND to Raspberry Pi Pin 6
- **Green** wire: switch EN to Raspberry Pi Pin 11, GPIO17
- **Red** wire: switch 5V to Raspberry Pi Pin 2
- **Yellow** Micro USB cable: switch input to the white USB hub
- **Black** USB cable: switch USB-A output to the sensor
- No DuPont connector is offset by one pin
- No bare conductor, loose solder, or debris can cause a short circuit
- All cables have adequate strain relief

Step 10: Power On and Test

26. Reconnect power to the assembly and allow the Raspberry Pi to boot.
27. Log into the Raspberry Pi and activate the GPIO17 control used by the SPARROW software.

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Log into the Raspberry Pi

From a computer on the same network (connect via the Ethernet port on the white USB hub), open a terminal and connect to the Raspberry Pi over SSH:

```
ssh <username>@<pi-ip-address>
```

Enter the Raspberry Pi password when prompted.

Turn GPIO17 ON

Set GPIO17 as an output and drive it HIGH:

```
sudo pinctrl set 17 op dh
```

Observe the USB device connected to the 8086 switch. If this SPARROW wiring is configured as active-high, the device should receive power and turn on.

Verify the GPIO17 state

Read the current GPIO17 configuration:

```
sudo pinctrl get 17
```

Confirm that GPIO17 is shown as an output driven HIGH.

Turn GPIO17 OFF

Drive GPIO17 LOW:

```
sudo pinctrl set 17 op dl
```

Observe the connected USB device. With active-high SPARROW wiring, it should lose power and turn off.

Turn GPIO17 back ON

Drive GPIO17 HIGH again:

```
sudo pinctrl set 17 op dh
```

Confirm that the USB device powers on again. This provides a visible ON/OFF/ON check of GPIO17 and the USB power-switch path.

Expected result

Command	**GPIO17**	**Expected USB state***
sudo pinctrl set 17 op dh	HIGH	ON
sudo pinctrl set 17 op dl	LOW	OFF
sudo pinctrl set 17 op dh	HIGH	ON

*Expected USB state assumes the SPARROW switch control is wired/configured as active-high. If your hardware is configured differently, verify the switch behaviour before using this as the operational test.

Quick test sequence

For a quick visual check after logging in:

```
sudo pinctrl set 17 op dh
sleep 3
sudo pinctrl set 17 op dl
sleep 3
sudo pinctrl set 17 op dh
```


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The connected USB device should visibly cycle ON, OFF, and ON when the control is active-high.

28. Confirm that the USB switch powers the connected sensor.
29. Confirm that the sensor enters USB storage mode.
30. Confirm that the sensor storage is accessible from the SPARROW Mini or SPARROW unit.
31. Deactivate the switch and confirm that the sensor leaves storage mode or powers down as expected.

EXPECTED OPERATION

When activated, the 8086.net USB Power Switch causes the compatible sensor to enter USB storage mode and passes the stored data through the white USB hub to the SPARROW Mini or SPARROW unit.

Quick Troubleshooting

Problem	First check
Switch does not activate	Verify GPIO17 is connected to EN and that GND is common.
Sensor does not enter storage mode	Confirm the sensor supports storage mode over USB and the USB cable carries data.
Sensor storage is not visible	Check the yellow cable, white hub connection, and the black sensor cable.
Raspberry Pi does not start	Disconnect power immediately and recheck the 5V, GND, and GPIO pin positions.

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13.3.2 SD/MicroSD Storage Interfacing: Access sensor storage via SD/MicroSD card slot.

Use this optional assembly for sensors that store captured data on an SD or microSD card but do not provide a USB data port or cannot automatically enter USB mass-storage mode. The Badgerd SDWire3 provides a remotely controlled SD-card interface between the sensor and the SPARROW Mini. The sensor writes to the installed card during normal operation. When the SPARROW software requests access, the SDWire3 switches the card to the Raspberry Pi so stored files can be downloaded without removing the card from the sensor.

ASSEMBLY OUTCOME: The sensor and SPARROW Mini share access to the same SD or microSD card through the SDWire3. Only one side should access the card at a time.

Required Parts and Tools

- Raspberry Pi Zero 2 W or Raspberry Pi Zero 2 WH, assembled as described earlier in this guide
- **Badgerd SDWire3** board
- Compatible SD or microSD card for the selected sensor
- **SDWire3 SD-card extension ribbon** or adapter that matches the sensor card slot
- **USB-C to USB-A data cable**, shown in the photographs as the **RED** cable
- White USB hub connected to the Raspberry Pi Zero 2 W
- Sensor or camera trap with an SD or microSD card slot
- Appropriate cable gland, gasket, sealing hardware, or weatherproof sealant if the sensor housing must be modified
- Solar-powered camera trap or sensor, recommended for long-term unattended deployments

Case Modification and Weatherproofing

WARNING: The SD-card extension ribbon must pass from the sensor card slot to the SDWire3 outside the sensor housing. Some sensors do not provide enough clearance for the ribbon. Drilling, cutting, filing, enlarging an opening, modifying the card door, or sensor housing or replacing part of the housing may be required. These changes may void the manufacturer's warranty and may reduce the original weather or ingress-protection rating. After routing the ribbon, install suitable strain relief and reseal the opening using an appropriately rated cable gland, gasket, sealant, or equivalent method. Inspect and test the modified enclosure for water and dust resistance before field deployment. Do not deploy a modified sensor until the housing has been properly resealed and weatherproofed.

For long-term deployments, use a solar-powered camera trap or sensor whenever practical. Confirm that the sensor power system can support the planned capture schedule and that all external power and data entries remain sealed for the deployment environment.

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Connection Overview

Connection	From	To
SD interface	Sensor SD/microSD slot	SDWire3 SD-card extension ribbon or adapter
Storage media	SDWire3 card holder	Compatible SD or microSD card
USB data/control	SDWire3 USB-C port	White USB hub using the RED USB-C to USB-A cable
Host connection	White USB hub	Raspberry Pi Zero 2 W Micro-USB data port

Step 1: Power Down and Prepare the Equipment

1. Shut down the Raspberry Pi Zero 2 W completely.
2. Turn off the sensor and disconnect any external power source.
3. Disconnect the white USB hub from the Raspberry Pi before changing the SDWire3 connections.
4. Work on a clean, dry, non-conductive and static-safe surface.
5. Read the sensor manufacturer's instructions for removing and installing the SD or microSD card.

Step 2: Prepare the SDWire3 and Storage Card

6. Identify the SDWire3 storage-card holder, sensor-side SD interface, and USB-C host connector.
7. Insert the compatible SD or microSD card into the SDWire3 card holder in the orientation indicated by the board and card socket.
8. Do not force the card. Remove it and verify orientation if resistance is encountered.
9. Confirm that the card is fully seated before connecting the sensor-side ribbon.



Figure 7. Aligning the storage card with the SDWire3 card holder.

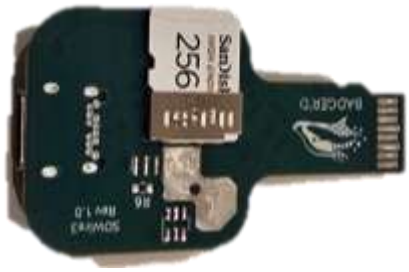


Figure 8. Storage card seated in the SDWire3 assembly.

Step 3: Connect the SD Extension to the Sensor

10. Open the sensor housing and locate the SD or microSD card slot.
11. Insert the sensor-side end of the SDWire3 extension into the sensor card slot using the same orientation required by a normal storage card.

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12. Route the ribbon along a smooth path. Do not sharply fold, twist, crease, puncture, or pinch the ribbon.
13. Leave enough slack for the housing to close without pulling on the sensor card slot or SDWire3 board.
14. If the housing cannot close around the ribbon, create a properly sized pass-through by modifying the housing and protect the ribbon edge from abrasion.
    IMPORTANT: Please follow the case-modification warning above and remember that **these changes may void the manufacturer's warranty and may reduce the original weather or ingress-protection rating.**



Figure 5. sensor-side SD extension components.



Figure 6. SD-card extension prepared for assembly.

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Step 4: Assemble the SDWire3 and SD Card Extension

15. Place the Badgerd SDWire3, microSD-to-SD adapter, and SD extension ribbon on a clean, non-conductive surface.
16. Align the SDWire3 SD-card interface with the slot in the microSD-to-SD adapter, as shown in Figure 9. Make sure the contacts are correctly oriented before inserting the board.
17. Carefully slide the SDWire3 into the microSD-to-SD adapter until fully seated. Do not force the connection.
18. Align the other side of the microSD-to-SD adapter with the SD extension ribbon assembly, as shown in Figure 10.
19. Insert the microSD-to-SD adapter into the SD extension connector until the adapter is fully seated, as shown in Figures 11 and 12.
20. Check the completed assembly. The connection should now follow this order:
21. Badgerd SDWire3 -> microSD-to-SD adapter -> SD extension ribbon -> sensor SD card slot
22. Route the ribbon toward the sensor without sharply bending, folding, twisting, or pinching the ribbon cable.
23. Confirm that every connection is fully seated before powering the sensor or connecting the SDWire3 to the SPARROW Mini.
    IMPORTANT: <u>Do not force any of the SD connections</u>. If a component does not slide into place normally, remove the component and check the orientation before trying again.



Figure 9. SDWire3 *aligned with the microSD to SD card adaptor slot.*



Figure 10. SDWire3 *is inserted into the microSD to SD card adaptor slot and aligned with the SD extension to sensor.*



Figure 11. Ribbon routed from the sensor toward the SDWire3.



Figure 12. SDWire3 *is inserted into the microSD to SD card adaptor and inserted into the ribbon routed to sensor.*

Step 5: Connect the SDWire3 to the White USB Hub

24. Connect the USB-C end of the **RED** USB-C to USB-A data cable to the USB-C port on the SDWire3 board.
25. Connect the USB-A end of the **RED** cable to an available port on the white USB hub.
26. Confirm that both connectors are fully seated.

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27. Secure the SDWire3 and cable so movement cannot pull on the USB-C connector, SD ribbon, or sensor card slot.
28. The **red** cable is used in the photographs to make this connection easy to identify. Any compatible USB-C to USB-A data cable may be used.



Figure 13. Red USB-C cable connected to the SDWire3 host port.



Figure 14. Red USB-A connector attached to the white USB hub.

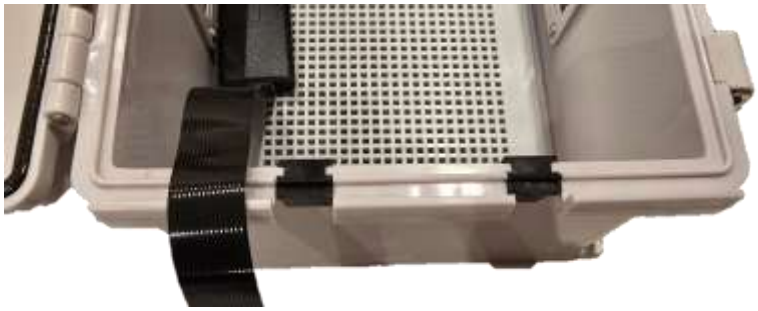


Figure 15. Complete SDWire3 path from sensor storage to the SPARROW Mini USB hub.
***IMPORTANT:** Modification of the Sparrow Mini housing might be necessary to accommodate proper ribbon cable routing. Remember that **these changes may reduce the original weather or ingress-protection rating**. After routing the ribbon, install suitable strain relief and reseal the opening using an appropriately rated cable gland, gasket, sealant, or equivalent weatherproofing method. Inspect and test the modified enclosure for water and dust resistance before field deployment. Do not deploy a modified unit until the housing has been properly resealed and weatherproofed.*

Step 6: Restore Power and Verify the Hardware

29. Reconnect the white USB hub to the Raspberry Pi Zero 2 W.
30. Reconnect power to the SPARROW Mini and allow the Raspberry Pi to boot.

(OPTIONAL) From a computer connected to the same network, open a terminal and connect to the Raspberry Pi through SSH:

```
ssh sparrow@<RASPBERRY-PI-IP-ADDRESS>
```

Replace <RASPBERRY-PI-IP-ADDRESS> with the IP address assigned to the Raspberry Pi, then enter the password when prompted.

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31. Restore power to the sensor.
32. (OPTIONAL) Confirm that the SDWire3 is detected by the Raspberry Pi.

List the connected SDWire devices:

```
sudo sdwire list
```

Confirm that the SDWire3 appears with a serial number or device identifier.
As an additional USB connection check, run:

```
lsusb
```

CHECKPOINT: Disconnecting and reconnecting the SDWire3 should cause the corresponding USB device to disappear and reappear in the lsusb output.

If the SDWire3 is not detected:

- Check the USB-C connection at the SDWire3.
- Check the USB-A connection at the white USB hub.
- Confirm that the hub is connected to the Raspberry Pi USB data port.
- Confirm that the red cable supports USB data and is not a charge-only cable.
- Try another available port on the white USB hub.

33. Confirm that the sensor can access the storage card while the SDWire3 is switched to the sensor side.

Set the SDWire3 so the sensor has access to the storage card:

```
sudo sdwire switch target
```

The equivalent DUT command may also be used:

```
sudo sdwire switch dut
```

If multiple SDWire devices are connected, specify the required serial number:

```
sudo sdwire switch -s <SDWIRE-SERIAL-NUMBER> target
```

Allow the sensor a few moments to recognize the card.
Confirm that the sensor can save a test image, recording, or sensor-data file.

34. Switch access to the Raspberry Pi only when the sensor is no longer writing to the card.

Confirm that the sensor is no longer writing to the SD card. Stop recording or capture activity if the sensor provides that option.
Disconnect the card from both sides:

```
sudo sdwire switch off
```

Assign the card to the Raspberry Pi:

```
sudo sdwire switch host
```


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IMPORTANT: Never switch the SD card while the sensor is writing. Interrupting a write operation may damage files or corrupt the filesystem.

35. Confirm that the card appears on the Raspberry Pi and that the expected image, audio, or sensor files can be read.

List available storage devices and filesystems:

```
lsblk -f
```

Identify the newly connected SD-card partition. It may appear as /dev/sda1, but the device name can vary.
If the new device is difficult to identify, inspect recent kernel messages:

```
sudo dmesg | tail -30
```

Create the mount point:

```
sudo mkdir -p /mnt/sensor
```

Mount the partition reported by lsblk -f:

```
sudo mount /dev/<SENSOR-PARTITION> /mnt/sensor
```

Example, only when the identified partition is /dev/sda1:

```
sudo mount /dev/sda1 /mnt/sensor
```

List and inspect the stored files:

```
ls -lah /mnt/sensor

find /mnt/sensor -maxdepth 3 -type f | head -50
```

VERIFY BEFORE MOUNTING: Always confirm the correct device name with lsblk -f. Mounting or modifying the wrong device may affect another connected drive or the Raspberry Pi system storage.

36. Safely unmount the card on the Raspberry Pi before returning access to the sensor.

After SPARROW finishes reading or copying the files, flush pending filesystem operations:

```
sync
```

Unmount the sensor card:

```
sudo umount /mnt/sensor
```

Confirm that /mnt/sensor no longer appears as a mount point:

```
lsblk -f
```

If the card reports that the device is busy:

```
cd ~

sudo fuser -vm /mnt/sensor
```


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Close or stop the process using the card, then run the unmount command again.

```
sudo umount /mnt/sensor
```

After a successful unmount, disconnect the card from the Raspberry Pi side:

```
sudo sdwire switch off
```

IMPORTANT: Do not return the card to the sensor until sudo umount /mnt/sensor completes successfully.

37. Return the SDWire3 to sensor access and confirm that the sensor resumes normal recording.

Assign the card back to the sensor:

```
sudo sdwire switch target
```

The equivalent DUT command may also be used:

```
sudo sdwire switch dut
```

Allow the sensor a few moments to recognize the card.
Confirm that the sensor resumes normal operation and can save a new test capture or recording.

IMPORTANT: Never allow the sensor and Raspberry Pi to write to the card at the same time. Do not switch ownership while either side is writing data as this may result in lost or corrupted data. Complete all transfers, flush pending writes, and unmount the storage on the Raspberry Pi before switching the card back to the sensor. Loss of power or incorrect switching during a write operation may corrupt the file system or stored data.

Final Inspection Checklist

- The storage card is fully seated in the SDWire3.
- The SD extension is fully inserted in the sensor card slot.
- The ribbon is not folded, pinched, twisted, or under tension.

The RED USB-C to USB-A cable connects the SDWire3 to the white USB hub.

- The white USB hub remains securely connected to the Raspberry Pi Zero 2 W.
- The SDWire3 board, ribbon, and cables have strain relief.
- Any modified housing opening has been sealed and weatherproofed.
- The sensor can record when card access is assigned to the sensor.
- The Raspberry Pi can read the card when access is assigned to the SPARROW Mini.
- The card is safely unmounted before access is returned to the sensor.

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Quick Troubleshooting

Problem	First check
SDWire3 is not detected	Check the RED USB-C to USB-A data cable, the white hub connection, and Raspberry Pi USB detection.
Card is not visible to the Raspberry Pi	Confirm that access has been switched to the host side and the card is fully seated.
Sensor cannot record	Return card access to the sensor side and confirm the sensor-side extension is inserted correctly.
Files or card appear corrupted	Stop using the card, check switching and unmount procedures, and verify that access was not changed during a write operation.
Water or dust can enter the housing	Do not deploy. Reseal the modified pass-through and verify the enclosure before field use.

14. Quick Troubleshooting

Problem	First check
Status page will not open	Make sure the phone/laptop is connected to the SPARROW Mini hotspot, then retry http://10.42.0.1:8080.
No XBee link	Confirm RELAY mode, antenna connection, compatible XBee 868/900 radios, and run Configure XBee.
No cloud upload	Confirm DIRECT mode, internet access, server details, API key, and onboarding.
No detections are being sent	Check detector status, blank filtering, and the backlog on the status page.
Setup needs to be repaired	Re-run the installer to re-apply the expected software configuration.
Hub XBee was replaced / link worked before, now dead	The node is pinned to the old hub chip's address. Open Settings → Rediscover hub and press Rediscover — it finds SPARROW_MASTER again and pins the new address.

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Wi-Fi/IP camera images not arriving	Confirm the camera is joined to the SPARROW Mini hotspot, FTP server = 10.42.0.1, port 21, user/password match, and address has no leading zeros (10.42.0.1, not 10.42.0.01). Check the Push receivers card shows FTP/SMTP running.
Camera connects but transfer fails mid-upload	Switch the camera's FTP transfer mode to Passive. Leave "Disable plain FTP" off (the node uses plain FTP, not FTPS).
Disk filling up / node slowing down	The camera is uploading video (.mp4) or .txt files, which are never sent and pile up. Turn off video/clip upload on the camera (stills only).
Status page shows "no signal" but images still send	Normal during a transfer — the radio is busy sending, so the heartbeat can't run ("radio busy"). It clears between cycles.
Hotspot (SSID) missing after a reboot	The Pi's Wi-Fi sometimes fails to load at boot. Reboot the node; if it's still missing, re-run the installer (it now waits for Wi-Fi and reloads the driver).
A setting I changed hasn't taken effect	Changes apply at the next capture cycle (up to launcher.loop_sleep_sec, default 30 min), not instantly. Wait one cycle.
Same image sent repeatedly / files look too big	Check the Backlog count on the status page. Confirm image compression is on and max_dim is set (≈800). A stuck backlog usually means the hub isn't acknowledging — check the hub.
Backlog not draining (relay mode)	The node is sending but the hub isn't replying (GO/OK). Verify the hub is powered, in range, its sparrow_service is running, and only one process holds its radio serial port.
Audio not being captured or sent	Check the Audio detector card is enabled, a mic/recorder is connected, and blank filtering isn't dropping everything (Settings → Blank frames).
Onboarding key rejected (direct mode)	The email must match the one the key was issued to, the key is one-shot (already consumed?), and the endpoint must be reachable.

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Not sure what a setting does	Open the Settings guide (/help, linked at the bottom of the status page) — it explains every setting and each Settings-menu action.

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